Exercise of Supervised Learning: Advanced Risk Minimization Part 3

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Exercise 3: Connection between between MLE and ERM

Suppose we are facing a regression task, i.e., $\mathcal{Y} = \mathbb{R}$, and the feature space $\mathcal{X} \subseteq \mathbb{R}^p$. Let us assume that the relationship between the features and labels is specified by

$$y = m^{-1}(m(f_{\text{true}}(\mathbf{x})) + \epsilon),$$

where $m : \mathbb{R} \to \mathbb{R}$ is a continuous strictly monotone function with m^{-1} being its inverse function, and the errors are Gaussian, i.e., $\epsilon \sim \mathcal{N}(0, \sigma^2)$. In particular, for the data points $(\mathbf{x}^{(1)}, y^{(1)}, \dots, \mathbf{x}^{(n)}, y^{(n)})$ it holds that

$$y^{(i)} = m^{-1}(m(f_{\text{true}}(\mathbf{x}^{(i)})) + \epsilon^{(i)}),$$

where $\epsilon^{(1)}, \dots, \epsilon^{(n)}$ are i.i.d. with distribution $\mathcal{N}(0, \sigma^2)$.
Disclaimer: We assume in the following that $m(y)$ and $m(f(\mathbf{x}))$ is well-defined for any $y \in \mathcal{Y}, f \in \mathcal{H}$ and $\mathbf{x} \in \mathcal{X}$.
(a) How can we transform the labels $y^{(1)}, \dots, y^{(n)}$ to "new" labels $z^{(1)}, \dots, z^{(n)}$ such that $z^{(i)} \mid \mathbf{x}$ is normally distributed? What are the parameters of this normal distribution?

Solution: Question (a)

Goal: transform $y^{(i)} \rightsquigarrow z^{(i)}$ such that $z^{(i)} \mid \mathbf{x}^{(i)}$ is Gaussian.

- Apply m: $y^{(i)} = m^{-1}(m(f_{\text{true}}(\mathbf{x}^{(i)})) + \epsilon^{(i)}) \implies m(y^{(i)}) = m(f_{\text{true}}(\mathbf{x}^{(i)})) + \epsilon^{(i)}$.
- Set $z^{(i)} := m(y^{(i)})$.
- Condition on $\mathbf{x}^{(i)}$: $m(f_{\text{true}}(\mathbf{x}^{(i)}))$ fixed, $\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2) \Rightarrow z^{(i)} \mid \mathbf{x}^{(i)}$ shifted Gaussian.
- $\implies z^{(i)} \mid \mathbf{x}^{(i)} \sim \mathcal{N}(m(f_{\text{true}}(\mathbf{x}^{(i)})), \sigma^2)$.

Solution: Question (a)

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Solution: Question (a)

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Question (b)

Assume that the hypothesis space is
$\mathcal{H} = \{f(\cdot, \boldsymbol{\theta}) : \mathcal{X} \to \mathbb{R} \mid f(\cdot|\boldsymbol{\theta})$ belongs to a certain functional family parameterized by $\boldsymbol{\theta} \in \Theta\}$,
where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_d))$ is a parameter vector, which is an element of a **parameter space** Θ. Based on your findings (a), establish a relationship between minimizing the negative log-likelihood for $(\mathbf{x}^{(1)}, z^{(1)}), \dots (\mathbf{x}^{(n)}, z^{(n)})$ and empirical loss minimizing over $\mathcal{H}$ of the generalized L2-loss function of Exercise sheet 1, i.e.,
$L(y, f(\mathbf{x}))) = (m(y) - m(f(\mathbf{x})))^2$.

Roadmap — two recipes to connect:

MLE on $(\mathbf{x}^{(i)}, z^{(i)})$.

1. Write the likelihood $\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{n} p(z^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\theta})$.
2. Take $-\log \to$ NLL $-\ell(\boldsymbol{\theta})$.

ERM with generalized L2.

1. Empirical risk $\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L(y^{(i)}, f(\mathbf{x}^{(i)}))$.
2. Plug in $L(y, f(\mathbf{x})) = (m(y) - m(f(\mathbf{x})))^2$.

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Solution to Question (b)

The likelihood for $(\mathbf{x}^{(1)}, z^{(1)}), \dots, (\mathbf{x}^{(n)}, z^{(n)})$:

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{n} p\left(z^{(i)} \mid f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}),\ \sigma^2\right)$$

Take $-\log \to$ NLL:

$\Rightarrow$ NLL $\propto$ empirical risk under generalized L2-loss.

MLE.

1. Write likelihood
2. Take $-\log \to$ NLL

ERM target.

1. Empirical risk $\sum_{i=1}^{n} L(y^{(i)}, f(\mathbf{x}^{(i)}))$
2. Plug in $(m(y) - m(f(\mathbf{x})))^2$

$\Rightarrow$ NLL $\equiv$ ERM (up to const.).

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The likelihood for $(\mathbf{x}^{(1)}, z^{(1)}), \ldots, (\mathbf{x}^{(n)}, z^{(n)})$:

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{n} p\left(z^{(i)} \mid f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}),\, \sigma^2\right)$$
$$\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[z^{(i)} - m(f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}))\right]^2\right).$$

Take $-\log \to$ NLL:

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MLE.

1. Write likelihood ✓
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The likelihood for $(\mathbf{x}^{(1)}, z^{(1)}), \ldots, (\mathbf{x}^{(n)}, z^{(n)})$:

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Take $-\log \rightarrow$ NLL:

$\Rightarrow$ NLL $\propto$ empirical risk under generalized L2-loss.

MLE.

1. Write likelihood $\checkmark$
2. Take $-\log \rightarrow$ NLL

ERM target.

1. Empirical risk $\sum_{i=1}^{n} L(y^{(i)}, f(\mathbf{x}^{(i)}))$
2. Plug in $(m(y) - m(f(\mathbf{x})))^2$

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Take $-\log \to$ NLL:

$$-\ell(\boldsymbol{\theta}) = -\log \mathcal{L}(\boldsymbol{\theta})$$

$\Rightarrow$ NLL $\propto$ empirical risk under generalized L2-loss.

MLE.

1. Write likelihood ✓
2. Take $-\log \to$ NLL ✓

ERM target.

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Take $-\log \to$ NLL:

$$-\ell(\boldsymbol{\theta}) = -\log \mathcal{L}(\boldsymbol{\theta})$$
$$\propto \sum_{i=1}^{n}\left[z^{(i)} - m(f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}))\right]^2$$
$$= \sum_{i=1}^{n}\left[m(y^{(i)}) - m(f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}))\right]^2.$$

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1. Empirical risk $\sum_{i=1}^{n} L(y^{(i)}, f(\mathbf{x}^{(i)}))$
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Question (c)

(c) In many practical applications, one often observed statisical property is that the label y given $\mathbf{x}$ follows a *log-normal distribution*. Note that we can obtain such a relationship by using $m(x) = \log(x)$ above. In the following we want to consider the conjecture of James D. Forbes, who conjectured in the year 1857 that the relationship between the air pressure y and the boiling point of water x is given by

$$y = \theta_1 \exp(\theta_2 x + \epsilon),$$

for some specific values $\theta_1 \in \mathbb{R}_+, \theta_2 \in \mathbb{R}$ and some error terms ϵ (of course, we assume that this error term is stochastic and normally distributed).
What would be a suitable hypothesis space $\mathcal{H}$ if this conjecture holds?

Solution to Question (c): setup & decomposition

- **Framework recap:** $y = m^{-1}(m(f(\mathbf{x})) + \epsilon)$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Two ingredients:
  - f — deterministic regression function, $f \in \mathcal{H}$ (what we choose);
  - m — fixed link function (*not* part of $\mathcal{H}$);
  - ϵ — Gaussian noise, applied *after* m.
- **What Q(c) is really asking:** pick the form of $f(\cdot \mid \theta)$ so that the framework reproduces Forbes' $y = \theta_1 \exp(\theta_2 x + \epsilon)$.
- **Why $m = \log$?** Q(c) says $y \mid \mathbf{x}$ log-normal $\iff \log y \mid \mathbf{x}$ Gaussian. Framework demands Gaussian residual after $m \implies m = \log$, $m^{-1} = \exp$.
- **Decompose:** with $m = \log$ the framework simplifies to

$$y = \exp(\log f(\mathbf{x}) + \epsilon) = f(\mathbf{x}) \cdot \exp(\epsilon),$$

i.e. deterministic signal $\times$ multiplicative noise. Match Forbes' to this shape next.

Solution to Question (c): setup & decomposition

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Solution to Question (c): algebra

Target shape (from setup): $y = f(\mathbf{x}) \cdot \exp(\epsilon)$.

- **Reshape Forbes' into the target shape:**

$$y = \theta_1 \exp(\theta_2 x + \epsilon)$$

- **Read off f & conclude:**

$$\mathcal{H} = \{f(x \mid \boldsymbol{\theta}) = \theta_1 \exp(\theta_2 x) : \boldsymbol{\theta} \in \Theta\}.$$

- **Sanity check (reparam.):** $\beta_0 := \log \theta_1,\ \beta_1 := \theta_2 \implies \log y = \beta_0 + \beta_1 x + \epsilon$ (linear Gaussian regression on $\log y$).

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$$\begin{aligned} y &= \theta_1 \exp(\theta_2 x + \epsilon) \\ &= \theta_1 \exp(\theta_2 x) \cdot \exp(\epsilon) \end{aligned}$$

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- **Sanity check (reparam.):** $\beta_0 := \log \theta_1,\ \beta_1 := \theta_2 \implies \log y = \beta_0 + \beta_1 x + \epsilon$ (linear Gaussian regression on $\log y$).

Solution to Question (c): alternative — linear in $\log y$

- **Apply $\log$ to both sides of Forbes':**

$$y = \theta_1 \exp(\theta_2 x + \epsilon) \iff \log y = \log \theta_1 + \theta_2 x + \epsilon.$$

$\implies$ ordinary linear regression on $z := \log y$.

- **Linear hypothesis space** (with $\mathbf{x} = (1, x)^\top \in \mathcal{X} = \{1\} \times \mathbb{R}$):

$$\mathcal{H} = \{ f(\mathbf{x} \mid \boldsymbol{\theta}) = \log(\theta_1)\, x_1 + \theta_2\, x_2 : \boldsymbol{\theta} \in \Theta \} = \{ \mathbf{x}^\top \boldsymbol{\theta} \}.$$

- **ERM = OLS:** $\hat{\theta} = (\log \hat{\theta}_1,\, \hat{\theta}_2)^\top = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{z}, \quad \boldsymbol{z} = (\log y^{(1)}, \ldots, \log y^{(n)})^\top.$

- **Closed form (simple-regression case):**

$$\hat{\theta}_2 = \frac{\sum_i (\mathbf{x}_2^{(i)} - \bar{x}_2)(\log y^{(i)} - \overline{\log y})}{\sum_i (\mathbf{x}_2^{(i)} - \bar{x}_2)^2}, \quad \hat{\theta}_1 = \exp\big(\overline{\log y} - \hat{\theta}_2\, \bar{x}_2\big).$$

Solution to Question (c): alternative — linear in $\log y$

- **Apply** $\log$ **to both sides of Forbes':**

$$y = \theta_1 \exp(\theta_2 x + \epsilon) \iff \log y = \log \theta_1 + \theta_2 x + \epsilon.$$

$\Longrightarrow$ ordinary linear regression on $z := \log y$.

- **Linear hypothesis space** (with $\mathbf{x} = (1, x)^\top \in \mathcal{X} = \{1\} \times \mathbb{R}$):

$$\mathcal{H} = \{f(\mathbf{x} \mid \boldsymbol{\theta}) = \log(\theta_1)\, x_1 + \theta_2\, x_2 : \boldsymbol{\theta} \in \Theta\} = \{\, \mathbf{x}^\top \boldsymbol{\theta}\,\}.$$

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Solution to Question (c): R code & fit on Forbes' data

```r
library(MASS); data(forbes); attach(forbes)
X <- cbind(rep(1, 17), bp); y <- pres

# (1) numeric ERM on Forbes' form
f <- function(x, theta)
  exp(theta[2]*x[2]) * theta[1] * x[1]
emp_risk <- function(theta)
  sum((log(y) - log(apply(X, 1, f, theta)))^2)
hat_theta <- optim(c(0.4, 0.5), emp_risk,
                   method = "L-BFGS-B",
                   lower = c(0, -Inf),
                   upper = c(Inf, Inf))$par
print(hat_theta)
## [1] 0.38050940 0.02059961

# (2) closed-form OLS on log(y)
hat_theta_2 <- cov(bp, log(pres)) / var(bp)
hat_theta_1 <- exp(mean(log(pres))
                   - hat_theta_2 * mean(bp))
```

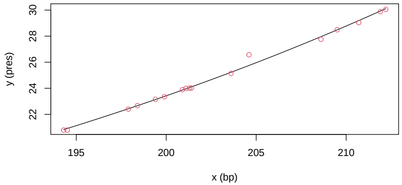

Fig. 1 — numeric ERM fit (red dots = data).

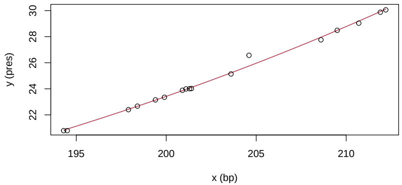

Fig. 2 — numeric (black) vs. closed-form OLS (red).

Solution to Question (c): observations

- Output: $\hat{\theta}_1 \approx 0.3805$, $\hat{\theta}_2 \approx 0.0206$.
  - $\hat{\theta}_2 \rightarrow$ log-slope $\approx 2\%$ per 1 °F.
  - $\hat{\theta}_1 \rightarrow$ scale $= f(0 \mid \boldsymbol{\theta})$.
- Numeric ERM $\equiv$ closed-form OLS $\rightarrow$ same $\hat{\theta}$.
- Fig. 1: ERM curve aligns with all 17 datapoints $\rightarrow$ Forbes' shape fits well.
- Fig. 2: ERM curve and OLS curve overlap on the same datapoints $\rightarrow$ identical fit.
- Takeaway: $m = \log \rightarrow$ nonlinear $\rightsquigarrow$ OLS, closed form for free.

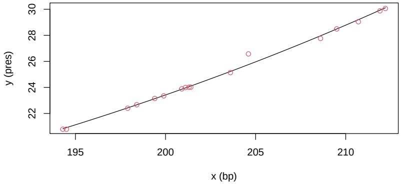

Fig. 1: ERM fit (black curve) on 17 datapoints (red dots).

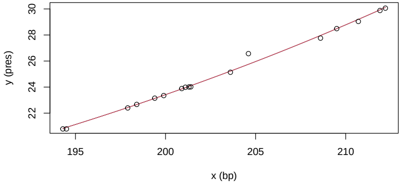

Fig. 2: ERM (black curve) overlaid with OLS (red curve), same 17

Solution to Question (c): observations

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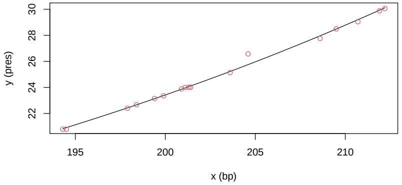

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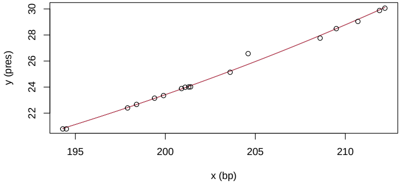

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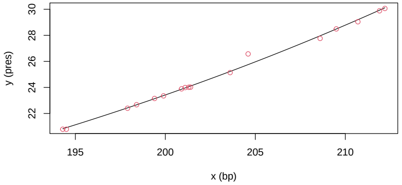

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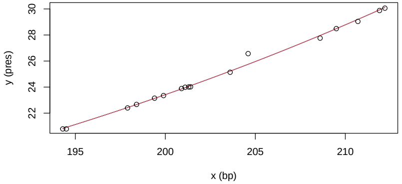

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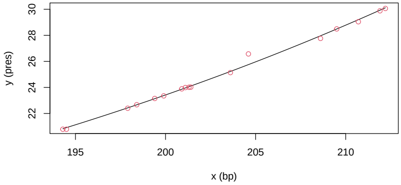

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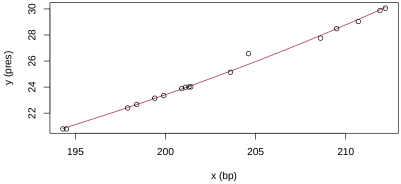

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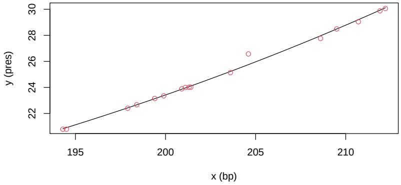

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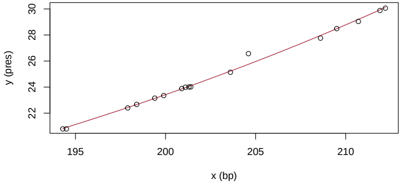

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Recap: link to the lecture material

Q	What the exercise teaches	Lecture material it builds on
(a)	The substitution $z^{(i)} = m(y^{(i)})$ exposes the hidden Gaussian: $z^{(i)} \mid \mathbf{x}^{(i)} \sim \mathcal{N}(m(f_{\text{true}}(\mathbf{x}^{(i)})), \sigma^2)$. The transformed labels obey the lecture's standard regression form.	Standard regression DGP $y = f_{\text{true}}(\mathbf{x}) + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The lecture only treats the identity link $m = \mathrm{id}$, where the Gaussian noise lives directly on the y-scale.
(b)	Define loss $:= -\log p$. Then minimizing NLL *is* empirical-risk minimization, so MLE and ERM are the same optimization. Plugging in the Gaussian density on $z^{(i)}$ produces the generalized L2 loss $L(y, f(\mathbf{x})) = (m(y) - m(f(\mathbf{x})))^2$.	The lecture's loss-as-negative-log-likelihood identity that fixes a loss for every noise model: Gaussian $\to$ L2, Laplace $\to$ L1, Bernoulli $\to$ log-loss. The exercise extends this catalogue with log-normal $\to$ generalized L2.
(c)	Choosing $\mathcal{H}$ = choosing the functional form of $f(\cdot \mid \boldsymbol{\theta})$. The hint $y \mid \mathbf{x}$ log-normal pins $m = \log$; matching Forbes' to the framework yields $\mathcal{H} = \{\theta_1 \exp(\theta_2 x) : \boldsymbol{\theta} \in \Theta\}$, fittable as OLS on $\log y$.	Hypothesis space as a parametric family $\{f_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta\}$; convex losses for exponential-family / GLM models. The derivation follows the same MLE-to-loss pattern as the log-cosh $\leftrightarrow$ cosh-MLE example shown in the lecture.